

Confidence Intervals Using the Paired t-Design

Statistics Worksheet · Grade 11–12 / AP Statistics

Name: _____

Date: _____

Learning Objectives

- Identify when a matched-pair (paired t) design is appropriate for two samples
- Compute the mean difference, standard deviation of differences, and degrees of freedom for a paired data set
- Construct and interpret a confidence interval for the population mean difference using the paired t formula

Problems

1. A researcher records the weight (in pounds) of 5 students before and after a 4-week diet program. The differences (Before minus After) are: 3, 5, 2, 4, 6. What is the mean difference?

$$\bar{d} = \frac{\sum d_i}{n}$$

2. Explain why the paired t-design is also called the 'matched pair design.' Give one real-world example where this design is appropriate.

3. A paired study compares two corn seed types (Regular vs. Kiln-Dried) on the same plots. The differences (Regular minus Kiln-Dried) for 6 plots are shown below. Complete the table by calculating each difference.

Plot	Regular (lb/acre)	Kiln-Dried (lb/acre)	Difference (d)
1	1903	2009	
2	1935	1915	
3	1910	2011	
4	2496	2180	
5	2108	1990	
6	1511	1535	



4. Using the six differences from Problem 3 (-106, 20, -101, 316, 118, -24), find the mean difference and the degrees of freedom for the paired t procedure.

$$\bar{d} = \frac{\sum d_i}{n}, \quad df = n - 1$$

5. The differences from a paired study of 11 corn plots have a mean difference of -33.727 and a standard deviation of differences of 66.171. Write the formula for the paired t confidence interval and identify what each variable represents.

$$\bar{d} \pm t^* \cdot \frac{s_d}{\sqrt{n}}$$

6. A paired t study has 11 pairs of observations. The standard deviation of differences is 66.171. Find the standard error of the mean difference.

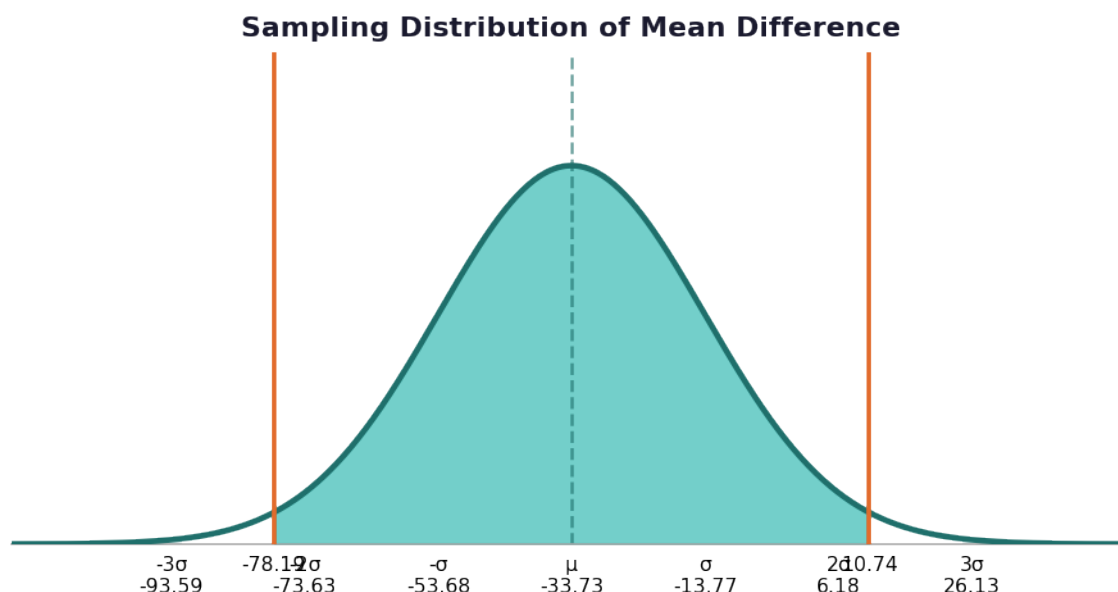
$$SE_{\bar{d}} = \frac{s_d}{\sqrt{n}}$$

7. Using the paired t distribution with 10 degrees of freedom at the 95% confidence level, the critical value t^* is 2.228. The mean difference is -33.727 and the standard error is 19.953. Construct the 95% confidence interval for the population mean difference.

$$\bar{d} \pm t^* \cdot \frac{s_d}{\sqrt{n}}$$

8. Based on the 95% confidence interval for the mean difference of regular versus kiln-dried corn yields found in Problem 7, can you conclude that there is a statistically significant difference between the two seed types? Explain your reasoning.





9. A physical therapist measures the range of motion (in degrees) of 8 patients' shoulders before and after treatment. The differences (After minus Before) are: 12, 9, 15, 7, 10, 13, 8, 6. The standard deviation of differences is 2.97. Construct a 95% confidence interval for the population mean improvement. Use $t^* = 2.365$ for $df = 7$.

$$\bar{d} \pm t^* \cdot \frac{s_d}{\sqrt{n}}$$

10. A study tests whether a new fertilizer increases tomato yield (pounds per plant). Eight matched plots are used; each plot is split, with one half receiving the new fertilizer and the other half receiving the standard fertilizer. The data are shown below. At the 99% confidence level (use $t^* = 3.499$, $df = 7$), construct the confidence interval for the mean difference (New minus Standard) and state whether the new fertilizer significantly increases yield.

Plot	New Fertilizer	Standard Fertilizer	Difference (d)	d ²
1	12.1	10.5	1.6	2.56
2	11.4	9.8	1.6	2.56
3	13.2	11.0	2.2	4.84
4	10.9	10.1	0.8	0.64
5	12.7	11.5	1.2	1.44



Plot	New Fertilizer	Standard Fertilizer	Difference (d)	d ²
6	14.0	12.3	1.7	2.89
7	11.8	10.2	1.6	2.56
8	13.5	11.2	2.3	5.29



Confidence Intervals Using the Paired t-Design — Answer Key

Statistics Worksheet · Grade 11–12 / AP Statistics

Answer Key

1. Answer: Mean difference = 4

- List the differences: 3, 5, 2, 4, 6
- Sum of differences = $3 + 5 + 2 + 4 + 6 = 20$
- $n = 5$
- Mean difference = $20 / 5 = 4$

2. Answer: Each observation in Sample 1 is paired with a specific observation in Sample 2 (e.g., same subject measured twice). Example: measuring blood pressure before and after medication on the same patients.

- In a matched pair design, pairs share a natural link (same individual, same plot of land, etc.).
- This controls for individual variability and focuses the analysis on the difference within each pair.
- Real-world example: testing the same students before and after a tutoring program.

3. Answer: Differences: -106, 20, -101, 316, 118, -24

Plot	Regular (lb/acre)	Kiln-Dried (lb/acre)	Difference (d)
1	1903	2009	-106
2	1935	1915	20
3	1910	2011	-101
4	2496	2180	316
5	2108	1990	118
6	1511	1535	-24

- Subtract Kiln-Dried from Regular for each plot.
- Plot 1: $1903 - 2009 = -106$
- Plot 2: $1935 - 1915 = 20$
- Plot 3: $1910 - 2011 = -101$
- Plot 4: $2496 - 2180 = 316$
- Plot 5: $2108 - 1990 = 118$
- Plot 6: $1511 - 1535 = -24$

4. Answer: Mean difference = 37.17, df = 5

- Sum of differences = $-106 + 20 + (-101) + 316 + 118 + (-24) = 223$





- Mean difference = $223 / 6 \approx 37.17$
- Degrees of freedom = $n - 1 = 6 - 1 = 5$

5. Answer: $\bar{d}$ is the mean of the differences; t^* is the critical value from the t-distribution; s_d is the standard deviation of the differences; n is the number of pairs.

- The formula is: $\bar{d} \pm t^* \times (s_d / \sqrt{n})$
- $\bar{d} = -33.727$ (mean of the paired differences)
- t^* = critical value from the t-table at the chosen confidence level with $df = n - 1$
- $s_d = 66.171$ (standard deviation of the paired differences)
- $n = 11$ (number of matched pairs)

6. Answer: SE ≈ 19.953

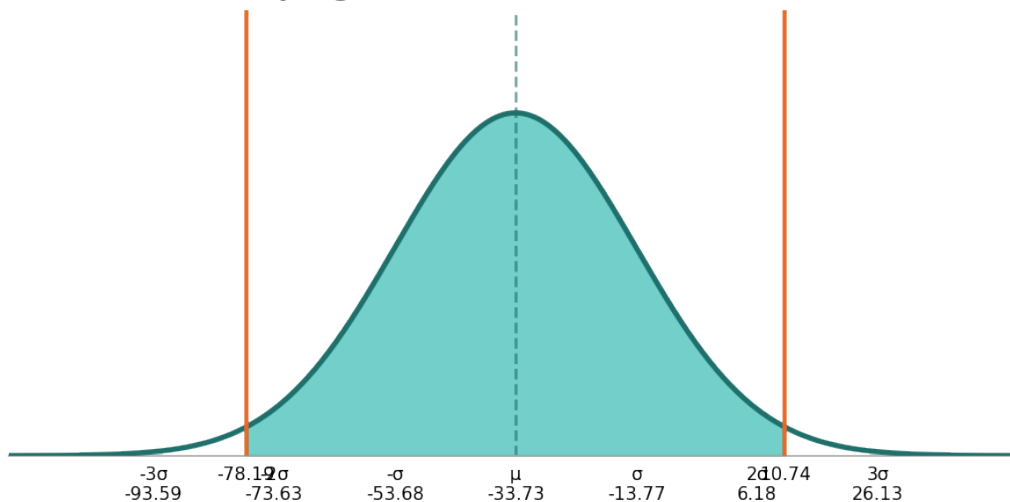
- $SE = s_d / \sqrt{n} = 66.171 / \sqrt{11}$
- $\sqrt{11} \approx 3.3166$
- $SE = 66.171 / 3.3166 \approx 19.953$

7. Answer: 95% CI: (-78.19, 10.74)

- Margin of error = $t^* \times SE = 2.228 \times 19.953 \approx 44.46$
- Lower bound = $-33.727 - 44.46 \approx -78.19$
- Upper bound = $-33.727 + 44.46 \approx 10.74$
- 95% Confidence Interval: (-78.19, 10.74)

8. Answer: No significant difference; zero is contained in the interval (-78.19, 10.74), so we cannot conclude the population mean difference is nonzero at the 95% confidence level.

Sampling Distribution of Mean Difference



- The 95% CI is (-78.19, 10.74).
- Since 0 is within this interval, it is plausible that the true mean difference is 0.
- Therefore, there is not enough evidence at the 95% confidence level to conclude a significant difference between regular and kiln-dried corn yields.

9. Answer: 95% CI: (7.518, 12.482)



- Sum of differences = $12 + 9 + 15 + 7 + 10 + 13 + 8 + 6 = 80$
- $n = 8$, so mean difference = $80 / 8 = 10$
- $SE = s_d / \sqrt{n} = 2.97 / \sqrt{8} = 2.97 / 2.8284 \approx 1.050$
- Margin of error = $t^* \times SE = 2.365 \times 1.050 \approx 2.483$
- Lower bound = $10 - 2.483 \approx 7.518$
- Upper bound = $10 + 2.483 \approx 12.482$
- 95% CI: (7.518, 12.482) — since 0 is not in the interval, the treatment shows a significant improvement.

10. Answer: Mean diff ≈ 1.625 ; $s_d \approx 0.492$; 99% CI: (0.993, 2.257) — Yes, significant increase.

- Differences: 1.6, 1.6, 2.2, 0.8, 1.2, 1.7, 1.6, 2.3
- Sum of differences = 13.0; $n = 8$; mean difference = $13.0 / 8 = 1.625$
- Sum of $d^2 = 2.56 + 2.56 + 4.84 + 0.64 + 1.44 + 2.89 + 2.56 + 5.29 = 22.78$
- $s_d^2 = [22.78 - 8 \times (1.625)^2] / (8 - 1) = [22.78 - 21.125] / 7 = 1.655 / 7 \approx 0.2364$
- $s_d = \sqrt{0.2364} \approx 0.4862$
- $SE = s_d / \sqrt{8} = 0.4862 / 2.8284 \approx 0.1719$
- Margin of error = $t^* \times SE = 3.499 \times 0.1719 \approx 0.602$
- Lower bound = $1.625 - 0.602 \approx 1.023$; Upper bound = $1.625 + 0.602 \approx 2.227$
- 99% CI: approximately (1.023, 2.227). Since 0 is NOT in the interval, the new fertilizer produces a significantly higher yield at the 99% confidence level.

